

CS 6501 — Week 5 Cheat Sheet: Liveness, Deadlock, and Minimally Invasive Prevention

Big Picture

Week 5 shifts from **safety** to **liveness**.

Safety ensures nothing bad happens (no collisions). Liveness ensures something good eventually happens (progress toward goal).

The core question:

How do we prevent deadlock in multi-robot systems in a minimally invasive way?

1. Liveness Definition

A robot is **live** if it continues to make progress toward its goal.

Freezing Robot Problem (FRP) occurs when:

$$|u_r|\Delta t < \epsilon \quad \wedge \quad x_r(t \in \Delta t) \notin x_g$$

Meaning:

- The robot is effectively not moving
- It has not reached its goal

Safety may still hold — but progress fails.

2. Deadlock

Deadlock is a special case of FRP.

Types:

- **Single-agent deadlock** (e.g., stuck around obstacle)
- **Multi-agent deadlock** (symmetry at doorway/intersection)

Key insight: Deadlock often arises from **symmetry**.

3. DS-MPEPC (Single-Agent Deadlock Resolution)

Idea: Add a **terminal cost term** that activates only under deadlock.

Original cost:

$$J_i = \sum (J_{progress} + J_{action} + J_{collision})$$

Modified cost:

$$J_i = \sum (...) + J_{terminal}$$

$$J_{terminal} = -k(C_{ttg} \cdot C_{ttc})$$

Where:

- C_{ttg} = time-to-goal cost

- C_{ttc} = time-to-collision cost

If deadlock occurs:

$$C_{ttg} \rightarrow 1$$

This encourages choosing trajectories that increase time-to-collision (wider path).

Key property: minimally invasive. The terminal term only activates during deadlock.

4. Social Mini-Games (SMGs)

Occurs when preferred trajectories conflict:

$$\tilde{\Gamma}_i \cap \tilde{\Gamma}_j \neq \emptyset$$

Examples:

- Doorways
- Hallways
- Intersections

In symmetric cases:

- Equal goal distance
- Equal velocity
- Equal priority

No agent yields $\rightarrow$ deadlock.

Symmetry $\Rightarrow$ deadlock tendency.

5. Buffered Voronoi Cells (Zhou et al.)

Partition space dynamically:

$$V_i = \{p \in \mathbb{R}^2 \mid (p - \frac{p_i + p_j}{2})^T p_{ij} \leq 0\}$$

Buffered version accounts for robot size.

Guarantees safety.

BUT: At narrow passages (doorway SMG), Voronoi regions overlap $\rightarrow$ robots freeze.

6. CBF Perspective on Deadlock (Wang et al.)

Recall CBF constraint:

$$\dot{h}(x) \geq -\alpha h(x)$$

CBF function:

$$h(x) = \|\bar{p}\| - \frac{\bar{v}_r^2}{2(\alpha_i + \alpha_j)} - d$$

As symmetry increases:

$$\bar{v}_r \uparrow \Rightarrow h(x) \downarrow$$

Eventually: No feasible safe control exists $\rightarrow$ deadlock.

Safety constraints shrink feasible control set.

7. Right-Hand Rule (Deadlock Resolution)

Break symmetry by biasing motion.

Heuristic: rightmost agent proceeds first.

Implementation methods:

Voronoi shift (Zhou): Shift partition boundaries leftward.

CBF modification (Wang): Split constraint:

$$\dot{h}(x) \geq -k_L \alpha h(x) \quad \wedge \quad -k_R \alpha h(x)$$

Bias:

$$k_L > k_R$$

Robot perceives right side as more unsafe $\rightarrow$ moves left.

RL approach (SACADRL): Reward shaping:

- Higher reward for right-hand motion

8. Autonomous Intersection Management

Centralized priority assignment:

- Scheduling
- Reservation systems
- Auctions

Relaxed taxonomy:

- Communicative
- Cooperative
- Centralized

More invasive but simpler to enforce.

9. Minimally Invasive Deadlock Prevention

Resolution = react after deadlock.

Prevention = avoid deadlock before it forms.

Goal setting:

- Decentralized
- Non-communicative
- Non-cooperative
- Minimally invasive

Minimally invasive: Adjust **speed**, not trajectory.

Open research question:

How do we prevent multi-robot deadlock in the hardest taxonomy setting?

10. Liveness Sets

Analogous to safe sets — but for velocity.

Definition:

$$\mathcal{C}_\ell(t) \subseteq \mathbb{R}^k$$

Joint velocity must satisfy:

$$v_i - \zeta v_j \geq 0, \quad \forall i \neq j, \zeta \geq 2$$

Matrix form:

$$Av \geq 0$$

For 2 robots:

$$A = \begin{bmatrix} 1 & -\zeta \\ -\zeta & 1 \end{bmatrix}$$

Interpretation: Velocity space is partitioned into:

- Deadlock zone
- Two liveness regions

Can be enforced as CBF constraint.

This enables: **Formal liveness guarantees.**

Core Themes to Remember

- Safety $\neq$ Liveness
- Symmetry $\Rightarrow$ Deadlock
- Resolution vs Prevention
- Minimally invasive control = speed perturbation
- Liveness sets = velocity-space analog of safe sets

One-Sentence Summary

Week 5 formalizes liveness, analyzes deadlock through symmetry and CBF geometry, and introduces minimally invasive prevention via velocity-based liveness sets.